Econometrics I

Lecture 9b: Staggered Adoption and Modern Difference-in-Differences

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Chris Conlon

NYU Stern

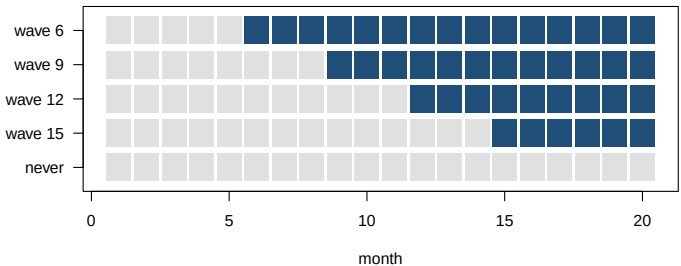
Fall 2026

Where we are

- Lecture 9a: one treatment date, a 2×2 comparison, parallel trends. Two-way FE got it exactly right.
- Most real rollouts are staggered: firms, stores or markets adopt at different dates.
- The default is to run the same two-way FE regression with D_{it} switching on at each unit's date.
  - What does that coefficient estimate when effects differ across cohorts or grow over time?
  - Which estimators fix it, and what do they assume?
  - Does it matter in published work?
- Plan: diagnose with a simulation where we know the answer, take the regression apart (FWL, then Goodman-Bacon), then rebuild it from clean 2×2s.

Running example: an algorithm rolled out city by city

- A delivery platform launches a new recommendation algorithm in 300 cities over 20 months: four waves of 60 cities (months 6, 9, 12 and 15), plus 60 cities that never get it. Outcome: log orders per user.
- Truth: nothing happens before launch. The effect ramps in over eight months, and earlier waves gain more (long-run effects 0.12, 0.09, 0.06, 0.04). The platform launched first where it expected the most.
- The true average effect over treated city-months is 0.061.



What two-way fixed effects estimates

Notation and targets

- G_i: the month unit i is first treated (its cohort); $G_i = \infty$ if never. Treatment is absorbing: $D_{it} = \mathbb{1}\{t \geq G_i\}$. Event time $e = t - G_i$.
- The building block is the cohort-by-period effect

$$\text{ATT}(g,t) = \mathbb{E}\left[\, Y_{it}(g) - Y_{it}(\infty) \mid G_i = g \,\right], \qquad t \geq g,$$

  where $Y_{it}(\infty)$ is the outcome if never treated.
- Summaries you might want, each a *weighted average* of the $\text{ATT}(g,t)$:
  - the overall ATT (every treated unit-period counts once);
  - the effect by cohort;
  - the effect by event time e: the dynamic path.
- Choose the summary before you choose the estimator. The trouble with two-way FE is that it chooses the weights for you.

The default regression

$$y_{it} = \alpha_i + \gamma_t + \beta \cdot D_{it} + \varepsilon_{it}$$

- On our rollout: $\widehat{\beta} = 0.024$ (SE 0.003). The truth is 0.061.
- Not noise: the standard error is small and every cohort's effect is positive.
- Not a failure of parallel trends either. The simulation satisfies parallel trends and no anticipation exactly.
- The problem is the estimand. To see it, apply Frisch–Waugh–Lovell.

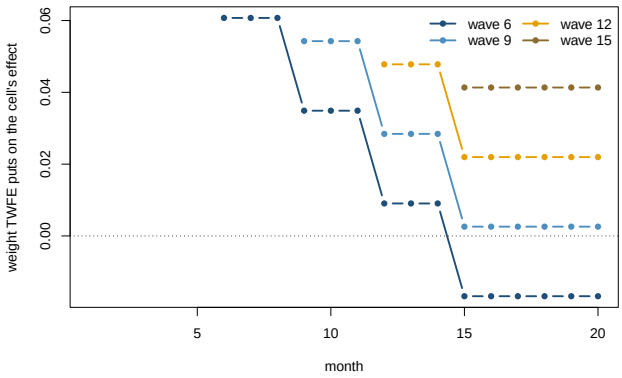
FWL: two-way FE is a weighted average with weights you did not choose

- Residualize D_{it} on unit and period effects: $\tilde{D}_{it} = D_{it} - \bar{D}_{i\cdot} - \bar{D}_{\cdot t} + \bar{D}$ (balanced panel). Then

$$\hat{\beta} = \frac{\sum_{it} \tilde{D}_{it}\, y_{it}}{\sum_{it} \tilde{D}_{it}^2} \quad \Rightarrow \quad \beta = \sum_{it:\, D_{it}=1} w_{it}\, \tau_{it}, \qquad w_{it} = \frac{\tilde{D}_{it}}{\sum_{js:\, D_{js}=1} \tilde{D}_{js}}.$$

- The weights sum to one, but they can be negative (de Chaisemartin and D'Haultfœuille 2020).
- Who gets a negative weight? A unit-period where $\bar{D}_{i\cdot}$ is high (an early adopter) and $\bar{D}_{\cdot t}$ is high (a late month, when most units are treated): $\tilde{D}_{it} < 0$.
- The weights depend only on the timing, not on outcomes. You can compute them before you look at y.

The weights in our rollout



- Each line is a wave's total weight per month. Wave 6 goes negative once most cities are treated.
- 14% of treated city-months get a negative weight, 8% of the absolute weight.
- Here the negative weights land on the *largest* effects: the early wave, late in its ramp-up. So the estimate is pulled down hard.
- With a constant effect, any weights summing to one would give the right answer.

Goodman-Bacon (2021): two-way FE as an average of 2×2s

- With timing groups k (earlier) and ℓ (later) and a never-treated group U, the two-way FE estimate is exactly a weighted average of every 2×2 DiD you can form:
  1. a timing group vs. the never-treated, over the whole panel;
  2. earlier vs. later, before the later group is treated: the later group is a *not-yet-treated* control;
  3. later vs. earlier, after the earlier group is treated: the earlier group is an *already-treated* control.
- Weights are larger for bigger groups and for pairs whose treatment share is near one half (timing in the middle of the panel).
- Comparison 3 is the "forbidden" one. Its control group's outcomes include its own treatment effects.

Why the already-treated comparison is not a treatment effect

- Later group ℓ vs. earlier group k, comparing the window after ℓ starts (POST) with the window in between (MID):

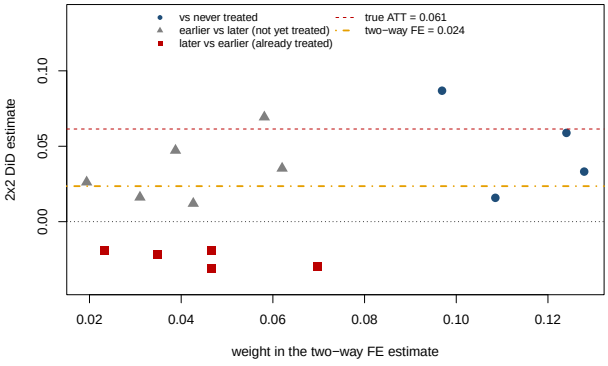
$$\hat{\tau}^{\ell}_{k\ell} = (\bar{y}^{POST}_{\ell} - \bar{y}^{MID}_{\ell}) - (\bar{y}^{POST}_{k} - \bar{y}^{MID}_{k}).$$

- Under parallel trends, the untreated trends cancel and what is left is

$$\underbrace{\bar{\tau}^{POST}_{\ell}}_{\text{a real effect}} - \underbrace{(\bar{\tau}^{POST}_{k} - \bar{\tau}^{MID}_{k})}_{\text{growth in the control group's effect}}.$$

- If effects are constant over time, the second term is zero and the comparison is fine.
- If the earlier group's effect is still *growing*, the second term is positive and the comparison is biased down. It can even be negative when every true effect is positive.

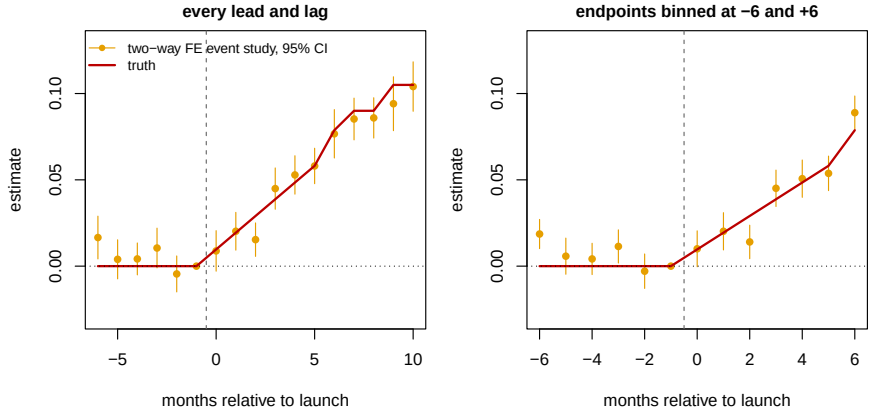
The decomposition in our rollout



Control group	N	Weight	Avg.
vs. never treated	4	0.46	0.047
vs. not yet treated	6	0.25	0.038
vs. already treated	6	0.29	-0.026

- All 16 comparisons, weighted, reproduce $\widehat{\beta} = 0.024$ to machine precision.
- The already-treated comparisons carry 29% of the weight and average -0.026.
- Even the clean comparisons are weighted oddly: the vs.-never 2×2s average 0.047, not 0.061.

Does a dynamic event study fix it?



- Mostly, but not cleanly. With cohorts whose paths differ, each lead or lag coefficient also picks up other cohorts' effects at *other* event times (Sun and Abraham 2021). Binning the endpoints, a common shortcut, makes it worse.
- A joint test of the leads rejects ($p = 0.04$ left, $p < 0.001$ right), though there is no pre-trend.

The fix: clean comparisons, chosen weights

The fix in one sentence

Estimate 2×2s that never use already-treated units as controls,
then average them with weights *you* choose.

Three ways to do it, which agree in our simulation:

1. Callaway and Sant'Anna (2021): estimate each ATT(g, t) as its own 2×2, then aggregate.
2. Sun and Abraham (2021): the same thing, written as one fully interacted regression.
3. Imputation (Borusyak, Jaravel and Spiess 2024; Gardner, Thakral, Tô and Yap 2024; Wooldridge 2025): fit the untreated outcome model on untreated observations only, predict $Y(0)$ for the treated, and average the gaps.

Callaway and Sant'Anna: ATT(g,t) one 2×2 at a time

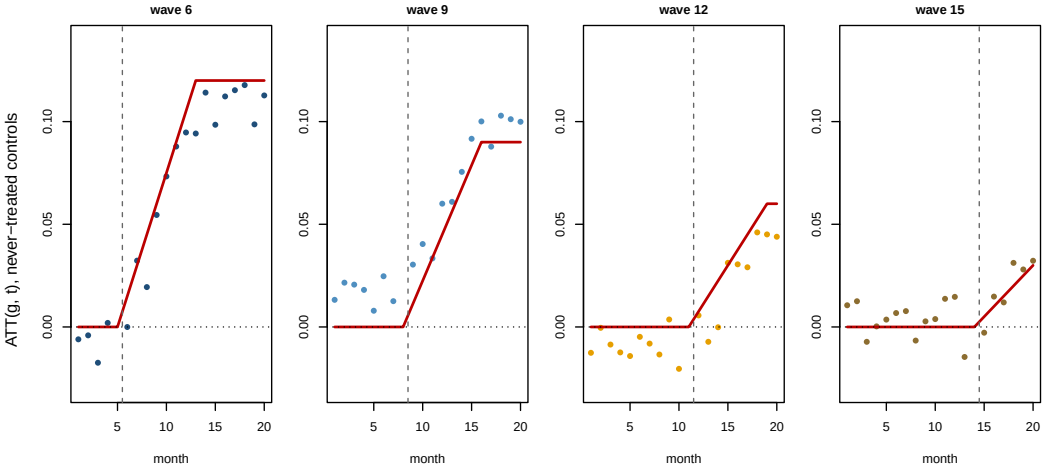
- For cohort g and period t, compare the change since the last pre-period $g-1$ with the change in a clean comparison group $\mathcal{C}$:

$$\widehat{\mathsf{ATT}}(g,t) = \Big(\bar{y}_{g,t} - \bar{y}_{g,g-1}\Big) - \Big(\bar{y}_{\mathcal{C},t} - \bar{y}_{\mathcal{C},g-1}\Big).$$

- Two choices of $\mathcal{C}$:
  - never-treated units ($G = \infty$);
  - not-yet-treated units ($G > t$). More controls, more power, but parallel trends must hold for more groups, and none of them may anticipate treatment.

- Assumptions: parallel trends between each cohort and its comparison group, and no anticipation. Covariates enter through outcome regression, weighting, or both (doubly robust, as in 9a).

ATT(g, t) in our rollout



One panel per wave; red is the truth. Each point is a 2×2 against the never-treated cities. Pre-period points are placebo 2×2s relative to month $g - 1$.

Aggregating ATT(g, t)

With cohort shares $P(G = g)$ as weights and $\mathcal{T}$ periods:

- Overall: average every post-period ATT(g, t), weighting by cohort size. Each treated unit-period counts once.
- By cohort: $\theta(g) = \frac{1}{\mathcal{T}-g+1} \sum_{t \geq g} \text{ATT}(g, t)$.
- By event time: the dynamic path

$$\theta(e) = \sum_{g} \mathbb{1}\{g + e \leq \mathcal{T}\}\, P(G = g \mid G + e \leq \mathcal{T})\, \text{ATT}(g, g + e).$$

- By calendar period: $\theta(t)$, averaging over cohorts treated by t.

Caveats:

- The cohorts behind $\theta(e)$ change with e. In our data only wave 6 is observed at $e = 14$.
- Long-horizon points describe early adopters, who may differ. Balance the panel in event time if you want a fixed composition.

Sun and Abraham: the same thing as a regression

- Interact cohort with event time, and use only clean controls:

$$y_{it} = \alpha_i + \gamma_t + \sum_{g} \sum_{e \neq -1} \delta_{g,e} \cdot \mathbb{1}\{G_i = g\} \cdot \mathbb{1}\{t - G_i = e\} + \varepsilon_{it},$$

  with never-treated (or last-treated) units as the comparison. Then average $\hat{\delta}_{g,e}$ across cohorts with cohort-share weights.

- Every treated cell except $e = -1$ has its own dummy, so the γ_t are estimated from clean controls only.

- With never-treated controls and no covariates, $\hat{\delta}_{g,e}$ is *numerically* the Callaway–Sant'Anna $\widehat{\text{ATT}}(g, g+e)$ with base period $g - 1$. In our simulation they agree to 12 decimal places.

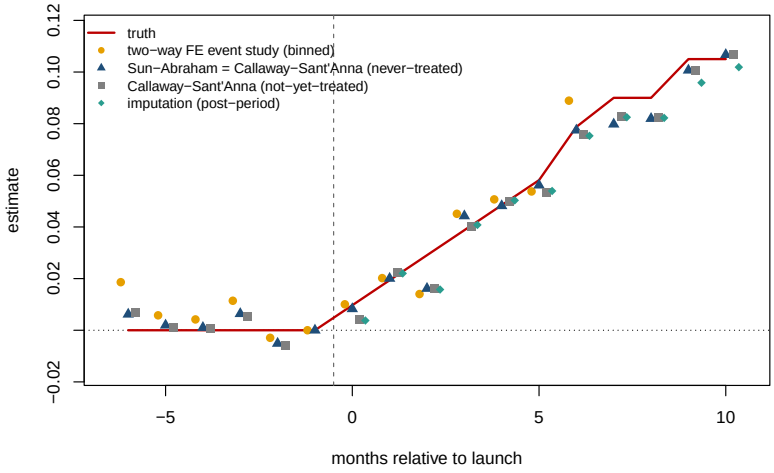
- In fixest: $\texttt{feols(y} \sim \texttt{sunab(cohort, month)} \mid \texttt{city + month)}$. Assignment 5 uses it.

Imputation

1. Fit $y_{it} = \alpha_i + \gamma_t + \varepsilon_{it}$ on the untreated observations only (never-treated, plus treated units before their date).
2. Predict $\widehat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\gamma}_t$ for every treated observation.
3. $\hat{\tau}_{it} = y_{it} - \widehat{Y}_{it}(0)$. Average with whatever weights you want: overall, by cohort, by event time.

- Borusyak, Jaravel and Spiess (2024) give the efficiency theory. Gardner et al.'s two-stage DiD ($\mathrm{did2s}$) is the same idea, and so is Wooldridge's (2025) extended two-way FE regression, a Mundlak-style version.
- Assumption: parallel trends in *all* periods, not just relative to $g - 1$. In exchange it uses every pre-period as a baseline, which buys precision when errors are not strongly serially correlated.
- Needs untreated observations for every unit and period it predicts: always-treated units drop out.

The horse race: event-time paths



Every clean estimator tracks the truth (red); the binned two-way FE event study drifts in the pre-period.

The horse race: one number

Estimator	Overall ATT	Std. error	Bias
Static two-way FE	0.024	(0.003)	-0.038
Sun–Abraham (sunab), never-treated controls	0.058	(0.005)	-0.003
Callaway–Sant'Anna, never-treated controls	0.058	(0.005)	-0.003
Callaway–Sant'Anna, not-yet-treated controls	0.058	(0.005)	-0.004
Imputation (Borusyak–Jaravel–Spiess)	0.057	(0.004)	-0.004
Truth: mean effect over treated city-months	0.061		

SEs for the hand-coded estimators: cluster bootstrap over cities (199 draws).

- Static two-way FE gives *less than half* the truth, with a small standard error.
- The clean estimators agree with each other. If they disagreed, an assumption would be doing work; find out which.
- Imputation is the most precise (SE 0.004 vs. 0.005), the payoff from its stronger assumption: it uses every pre-period, not just $g-1$.

Practice

Choosing the comparison group

- Never-treated. Cleanest, but ask why they never adopted. Never-adopters are often different (smaller, rural, uninterested), and parallel trends with them is a strong claim.
- Not-yet-treated. Units that will adopt soon are often more comparable. Requires no anticipation.

- No never-treated group: use the last cohort as the control until it is treated. Effects after that date are not identified without extrapolation.
- Always-treated units (adopted before the sample starts) cannot be a clean control and have no pre-period. Drop them. Two-way FE instead uses them as controls throughout.
  - In a variant of our rollout with 50 always-treated cities, 24% of treated rows get negative weights and two-way FE gives 0.015 (truth 0.061). Imputation without them gives 0.060.
- Anticipation: if units react δ periods early, use $g - 1 - \delta$ as the base period.

Pre-period plots are not comparable across estimators

- Regression event studies (two-way FE, Sun–Abraham) report every coefficient relative to $e = -1$: long differences on both sides.
- $\mathrm{did::att_gt}$ by default uses a varying base period: pre-period points are *short* differences (t vs. $t-1$), post-period points are long differences (t vs. $g-1$). A steady pre-trend then looks flat before and kinked after.
- Imputation's pre-trend test is a different object again: it regresses untreated outcomes on leads.
- So do not eyeball a "flat pre-period" across methods. For plots comparable to a regression event study, set $\mathrm{base_period} = \text{"universal"}$ (Roth 2026 WP, "Interpreting Event-Studies from Recent Difference-in-Differences Methods"). Our figures use the universal base.

Before you estimate anything

1. Plot the rollout. Which cohorts, how big, and when? (panelview.)
2. Count never-treated and always-treated units. Few never-treated units and many always-treated ones are where two-way FE goes wrong.
3. Compute the two-way FE weights. They depend only on timing. Report the share that is negative.
4. Run the Goodman-Bacon decomposition. How much weight sits on already-treated comparisons, and do those 2×2s disagree with the clean ones?
5. Estimate with a clean estimator. Report the event-time path *and* your chosen aggregate. Put two-way FE beside it.
6. If they differ, the decomposition tells you why. If they agree, say so: it is reassuring.

Does the TWFE Problem Matter in Practice? (Hull 2026)

- AI-assisted replication of 1,346 staggered-adoption specs from 108 papers (about half of the specs from top-five journals).
- Compare TWFE to DiD-averaging (Callaway–Sant'Anna) and imputation (Borusyak–Jaravel–Spiess).

	DiD-averaging	Imputation
Correlation with TWFE estimate	0.88	0.93
Sign flips	14.5%	10.8%
Sign flips with a significant difference	3.5%	2.2%
Modern SE relative to TWFE SE	+47%	+14%

- Negative weights are small in the median static spec: 8% of treated rows, 1.4% of absolute weight.
- The tradeoff: robustness to TWFE's implicit weighting vs. the precision of OLS.

Separating Noise from True Differences (Hull 2026)

- Hierarchical model: $\hat{\theta}_k \mid \theta_k \sim N(\theta_k, S_k)$ for each (TWFE, modern) pair; bivariate normal prior on θ_k.

Estimands	DiD-averaging	Imputation
Correlation of TWFE and modern estimands	0.91	0.98
Sign reversal	11.5%	5.6%
Significant TWFE estimate with the wrong sign	2.1%	0.6%

- When to worry: +10pp always-treated share $\Rightarrow$ +6.7pp sign flips (DiD-averaging; forbidden comparisons); +10pp never-treated share $\Rightarrow -2.1$pp (imputation).
- Flips are rarer after 2021 and when papers report both estimators. Caveat: these are *published* specifications.
- Run the modern estimator anyway (it's cheap), but old TWFE results are not presumptively wrong.

Our simulation vs. Hull's sample

- Our rollout was built to break two-way FE:
  - effects that grow for eight months;
  - larger effects for earlier waves;
  - a short panel in which most cities end up treated.

  There, 14% of treated rows carry negative weight; in Hull's median spec, 8%.
- The lesson is not "two-way FE is always badly wrong." It is: two-way FE is right under *constant* effects, and you can check cheaply how far from that you are.
- The diagnostics on the previous slides tell you which world you are in before a referee does.

Beyond binary, absorbing treatments (pointers)

- Treatments that switch on and off, or units whose treatment changes more than once: de Chaisemartin and D'Haultfœuille (2020, *AER*; 2026, *REStat*), which compare switchers with units whose treatment has not yet changed.
- Continuous treatment (dose: ad spend, a tax rate): Callaway, Goodman-Bacon and Sant'Anna (2024 WP). Comparing dose levels needs a stronger parallel-trends assumption than comparing treated with untreated.
- Sensitivity to violations of parallel trends: HonestDiD (Rambachan and Roth 2023; Session 8).
- Few treated units, long pre-period: synthetic control and synthetic DiD (Session 11).
- Survey: Roth, Sant'Anna, Bilinski and Poe (2023, *JoE*), "What's trending in difference-in-differences?"

In practice

```
# cohort g = first treated month; never-treated: g = 0 for did/didimputation, g = 10000 for sunab
fit <- feols(y ~ sunab(g_sa, month) | city + month, d, vcov = ~city)            # Sun-Abraham
summary(fit, agg = "ATT")                                                       # overall ATT
att <- did::att_gt(yname = "y", tname = "month", idname = "city", gname = "g", data = d,
                   control_group = "notyettreated", base_period = "universal")
did::aggte(att, type = "dynamic")                                               # also "simple", "group", "calendar"
didimputation::did_imputation(data = d, yname = "y", gname = "g", tname = "month", idname = "city",
                              horizon = TRUE, pretrends = -5:-1)                # imputation
bacondecomp::bacon(y ~ D, data = d, id_var = "city", time_var = "month")        # Goodman-Bacon weights
```

```
pf.panelview(d, unit="city", time="month", treat="D")                           # the rollout
fit = pf.event_study(d, yname="y", idname="city", tname="month", gname="g", estimator="did2s")
fit = pf.did2s(d, yname="y", first_stage="~ 0 | city + month",
               second_stage="~ i(rel, ref=-1.0)", treatment="D", cluster="city")
```

Recap

- With staggered timing and effects that change, static two-way FE is a weighted average of cohort-period effects with timing-driven weights. Some are negative, and the already-treated comparisons are not treatment effects.
- Dynamic two-way FE event studies are better but still mix cohorts, and binning makes it worse.
- Fix: clean 2×2s (never- or not-yet-treated controls), aggregated with weights you choose. Callaway–Sant'Anna, Sun–Abraham and imputation all do this; imputation trades a stronger assumption for precision.
- Before estimating: plot the rollout, count never- and always-treated units, compute the weights, decompose.
- In published work the damage is usually modest (Hull 2026). Run the modern estimator and report both.

Workbook time

Fall/L9 - Quasi Experiments/inclass_session9.Rmd, Part 2 (Python: inclass_session9_py.qmd):

1. Reproduce static two-way FE, sunab, hand-coded Callaway–Sant'Anna and imputation on the rollout, against the truth.
2. Make the effect constant across waves and over time. Which estimators change, and why?
3. Drop the never-treated cities. Which comparisons are left, and which event-time effects can you still estimate?